1. Describe the data pipeline in your own words. Where does the raw data come from, how is it processed, and what does the model ultimately predict?\

   The data comes from historical NBA stats used for training and live betting odds scraped from FanDuel for current games. The data is processed into win percentages, point differentials, and recent performance stats that the models can interpret. The XGBoost and neural network models predict win probabilities for each team and over/under totals for points scored.

2. The project uses two model types: XGBoost and a neural network. Based on what you know from class, what is one advantage of each approach?

   XGBoost provides clear interpretability through feature importance scores, showing which factors (like home court advantage or recent form) drive predictions, and it performs well on tabular sports data without requiring massive datasets, while Neutral Network can automatically discover complex, non-linear patterns and feature interactions without manual engineering — for example, finding how multiple factors combine in unexpected ways to affect game outcomes.

3. What does "expected value" mean in the context of sports betting predictions? How is it different from simply predicting who wins?

   Expected value calculates and determines the average amount of gain or loss of a bet based on the probability of winning and the odds. So if its positive, that means the bet is likely to be profitable over time while a negative suggests a loss.

4. What are two limitations or ethical concerns you would consider before deploying a tool like this in the real world?

   I think one would be the risk of gambling addiction since it may encourage repeated and impulsive betting. This tool may make betting seems logical and safe, while in real life, if some data can't be processed in the code, it may lead to financial ruin. Another concern would be this model being unreliable. Sports have so many variables, such as trades, injuries and more, so overtrusting predictions can lead to devastating losses, especially when certain stats are unavailable.